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APPROXIMATION OF ENTIRE FUNCTIONS OF EXPONENTIAL TYPE BY EXPONENTIAL SUMS*

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Abstract. Let K be a compact set in $\mathbb{R}^n$. For $1 \leqslant p \leqslant \infty$, the Bernstein space B_K^p is the Banach space of all functions $f \in L^p(\mathbb{R}^n)$ such that their Fourier transform in a distributional sense is supported on K. If $f \in B_K^p$, then f is continuous on $\mathbb{R}^n$ and has an extension onto the complex space $\mathbb{C}^n$ to an entire function of exponential type K. We study the approximation of functions in B_K^p by finite τ-periodic exponential sums of the form

$$\sum_m c_m \mathrm{e}^{2\pi \mathrm{i}(x,m)/\tau}$$

in the $L^p(\tau[-1/2,1/2]^n)$-norm as $\tau \to \infty$ when K is a polytope in $\mathbb{R}^n$.

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1 INTRODUCTION

The Fourier transform on $L^1(\mathbb{R}^n)$ is defined here by

$$\hat{f}(x) = \int_{\mathbb{R}^n} f(t)\mathrm{e}^{-\mathrm{i}(x,t)}\,\mathrm{d}t,$$

where $(x,t) = \sum_{k=1}^n x_k t_k$ denotes the inner product in $\mathbb{R}^n$. We normalize the inverse Fourier transform

$$\check{f}(\xi) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} f(t)\mathrm{e}^{\mathrm{i}(\xi,t)}\,\mathrm{d}t \tag{1.1}$$

so that the inversion formula $\widehat{(\check{f})} = f$ holds for suitable f. If $f \in L^p(\mathbb{R}^n)$ and $p > 2$, then the Fourier transform $\hat{f}$ is understood as a tempered distribution on $\mathbb{R}^n$.

Let K be a compact set in $\mathbb{R}^n$. For $1 \leqslant p \leqslant \infty$, the Bernstein space B_K^p is the closed subspace of $L^p(\mathbb{R}^n)$ consisting of all $f \in L^p(\mathbb{R}^n)$ such that $\hat{f}$ is supported on K. In another terminology, any $f \in B_K^p$ is

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bandlimited to K. The Banach space B_K^p is equipped with the norm

$$\|f\|_p = \Big(\int\limits_{\mathbb{R}^n} \big|f(x)\big|^p \,\mathrm{d}x \Big)^{1/p} \quad \text{if } 1 \leqslant p < \infty$$

and

$$\|f\|_\infty = \operatorname*{ess\,sup}_{\mathbb{R}^n} \big|f(x)\big| \quad \text{if } p = \infty.$$

Throughout this article, we consider only the cases where B_K^p is an infinite-dimensional Banach space. For this reason, we will assume that K is a convex compact body in $\mathbb{R}^n$. By a convex compact body we mean here a convex compact subset $K \subset \mathbb{R}^n$ which coincides with the closure of its inner point set $\operatorname{int}(K)$. By the Paley–Wiener–Schwartz theorem (see [4, p. 181]), any $f \in B_K^p$ is continuous on $\mathbb{R}^n$ and has an extension onto the complex space $\mathbb{C}^n$ to an entire function of exponential type K; this means that

$$\big|f(z)\big| \leqslant M\mathrm{e}^{H_K(-y)} \tag{1.2}$$

for all $z = x + \mathrm{i}y$, $x, y \in \mathbb{R}^n$, where M is a positive constant, and

$$H_K(\xi) = \max_{t \in K} (\xi, t), \quad \xi \in \mathbb{R}^n,$$

is the supporting function of K.

Let $\mathbb{Z}^n$ denote the lattice of integer vectors in $\mathbb{R}^n$, and let $\mathbb{T}^n = [-1/2, 1/2]^n$. We write $\mathbb{N}$ for the set of natural numbers. Fix $\tau > 0$. Since each $f \in B_K^p$ is a continuous function on $\mathbb{R}^n$, we have that the Fourier coefficients of f on $\tau\mathbb{T}^n$,

$$c_m(f, \tau) = \frac{1}{\tau^n} \int\limits_{\tau\mathbb{T}^n} f(u)\mathrm{e}^{-2\pi\mathrm{i}(m,u)/\tau} \,\mathrm{d}u, \quad m \in \mathbb{Z}^n,$$

are well defined. The Fourier series of $f \in B_K^p$ is formally given by

$$\sum_{m \in \mathbb{Z}^n} c_m(f, \tau)\mathrm{e}^{2\pi\mathrm{i}(m,x)/\tau}. \tag{1.3}$$

We will take in (1.3) only those terms whose Fourier transforms are supported on K. More precisely, set

$$f_\tau(x) = \sum_{m \in \mathbb{Z}^n \cap n_\tau K} c_m(f, \tau)\mathrm{e}^{2\pi\mathrm{i}(m,x)/\tau}, \tag{1.4}$$

where $n_\tau = [\tau/2\pi]_\mathbb{Z}$ is the integer part of $\tau/2\pi$, and $n_\tau K = \{n_\tau u\colon\ u \in K\}$. Since K is a compact set, f_τ is a trigonometric polynomial on $\mathbb{R}^n$. We also consider the following $(\tau\mathbb{T}^n)$-truncated version of f_τ:

$$\varphi_{f,\tau} = f_\tau \chi_{\tau\mathbb{T}^n}, \tag{1.5}$$

where $\chi_{\tau\mathbb{T}^n}$ is the characteristic (indicator) function of the set $\tau\mathbb{T}^n$.

Let now $K = [-\sigma, \sigma] \subset \mathbb{R}$, $\sigma > 0$, and set $B_\sigma^2 := B_{[-\sigma,\sigma]}^2$. In a recent paper, Martin [7] studied the approximation of $f \in B_\sigma^2$ by the sums of one variable,

$$\chi_\tau(z) \sum_{|m| \leqslant [\tau\sigma/2\pi]_\mathbb{Z}} c_m(f, \tau)\mathrm{e}^{2\pi\mathrm{i}mz/\tau}, \tag{1.6}$$

where $z \in \mathbb{C}$, and χ_τ is the characteristic function of the strip $\{z \in \mathbb{C}\colon |\operatorname{Re} z| \leqslant \tau/2\}$. Recall that a function f is called strictly bandlimited to $[-\sigma, \sigma]$ if it is bandlimited to $[-\varrho, \varrho]$ for some $0 \leqslant \varrho < \sigma$. It was proved in [7] that if $f \in B^2_\sigma$ and f is strictly bandlimited to $[-\sigma, \sigma]$, then (1.6) converges to f in the L^2 norm as $\tau \to \infty$ on $\mathbb{R}$ and on any horizontal line in $\mathbb{C}$. Approximations of $f \in B^2_\sigma$ by trigonometric polynomials similar to (1.6) were considered earlier by Gröchenig [2], who also gave a quantitative estimate of approximation in an appropriate L^2 norm. In [12], we studied the approximation of elements of $B^p_\sigma := B^p_{[-\sigma,\sigma]}$, $1 < p < \infty$, by the sums (1.5) of one variable. In the case of several variables, we investigated (1.5) for $f \in B^2_K$ (see [11]). Based on the Hilbert space techniques, we presented a simple proof of the approximation statement for an arbitrary Jordan-measurable compact subset $K \subset \mathbb{R}^n$.

In the present paper, we give another approach to the approximation of $f \in B^p_K$ by (1.5) in the more general case of $p \in (1, \infty)$. By Fefferman's theorem on spherical multipliers (see, for example, [15, p. 66]), it follows that if $p \neq 2$, then the trigonometric system on $\mathbb{T}^n$ is not a basis in $L^p(\mathbb{T}^n)$ for spherical summation. Therefore, we are forced to consider the sums (1.5) for some particular classes of K. In particular, these classes should not contain any ball in $\mathbb{R}^n$. In this paper, we study the case where K is a polytope. We recall that by an *n-polytope* in $\mathbb{R}^n$ we mean any intersection P of a finite number of half-spaces $L(a, \omega) = \{x \in \mathbb{R}^n\colon (x, a) \leqslant \omega\}$, $a \in \mathbb{R}^n$, $\omega \in \mathbb{R}$, such that P has a nonempty interior $\operatorname{int}(P)$.

Theorem 1. *Suppose that $f \in B^p_K$ for $1 < p < \infty$. If $K \subset \mathbb{R}^n$ is an n-polytope, then*

$$\lim_{\tau\to\infty} \|f - \varphi_{f,\tau}\|_{L^p(\mathbb{R}^n)} = 0. \tag{1.7}$$

Until recently, the approximation of bandlimited functions by trigonometric polynomials was most completely investigated in the case $p = \infty$. Lewitan [6] was the first to define the trigonometric polynomials

$$\tilde{f}_\tau(x) := \sum_{k\in\mathbb{Z}} f(x + k\tau)\frac{\sin^2(\frac{x}{\tau} + k)}{(\frac{x}{\tau} + k)^2}, \quad \tau > 0,\ x \in \mathbb{R}. \tag{1.8}$$

It was proved in [6] that if $f \in B^\infty_\sigma$, then $\tilde{f}_\tau \to f$ as $\tau \to \infty$ uniformly on compact subsets of $\mathbb{R}$. Krein [5] obtained a new proof of this statement, investigated the degree of approximation, and considered several applications. Hörmander [3] considered trigonometric polynomials that are more general than (1.8) (for a more detailed history and references, see [1, pp. 146–152]). For $f \in B^p_\sigma$ and $1 \leqslant p < \infty$, Schmeisser [13] proved convergence theorems for Lewitan (or Lewitan–Krein–Hörmander) polynomials (1.8) with respect to L^p norms on lines in $\mathbb{C}$ parallel to $\mathbb{R}$. The approximation by polynomials of several variables similar to (1.8) was also studied in the case $f \in B^\infty_K$ (for details, see [10] and the references therein).

For our final theorem, we return to the polynomials (1.5). As in the case of Lewitan–Krein–Hörmander polynomials, it turns out that (1.5) also converges uniformly on compact subsets of $\mathbb{R}^n$.

Theorem 2. *Suppose that $f \in B^p_K$, where $1 < p < \infty$. If K is an n-polytope, then $\lim_{\tau\to\infty}(f - \varphi_{f,\tau}) = 0$ uniformly on compact subsets of $\mathbb{R}^n$.*

2 PRELIMINARIES

Let us start by introducing some terminology and notation. For $x \in \mathbb{R}^n$, we denote by $|x|_p$ the l_p norm on $\mathbb{R}^n$, i.e.,

$$|x|_p = \Big(\sum_k |x_k|^p\Big)^{1/p} \quad \text{if } p \in [1, \infty)$$

and $|x|_\infty = \max_k |x_k|$. Set

$$\mathbb{U}^n = \big\{x \in \mathbb{R}^n\colon |x|_2 \leqslant 1\big\} \quad \text{and} \quad \mathbb{Q}^n = \big\{x \in \mathbb{R}^n\colon |x|_\infty \leqslant 1\big\}.$$

If A and B are two sets in $\mathbb{R}^n$, we write $A + B$ for the set of all sums $a + b$, $a \in A$, $b \in B$. In particular, if $A = \{a\}$, $a \in \mathbb{R}^n$, we write simply $a + B$. For $\delta \in \mathbb{R} \setminus \{0\}$, we denote by δA the set $\{\delta a\colon\ a \in A\}$. Let K be a compact subset of $\mathbb{R}^n$. The Lebesgue measure (in $\mathbb{R}^n$) of K will be denoted by $m_n(K)$. Set

$$r_K = \inf\bigl\{|x|_2\colon\ x \in \mathbb{R}^n \setminus K\bigr\} \tag{2.1}$$

and

$$R_K = \max\bigl\{|x|_2\colon\ x \in K\bigr\}. \tag{2.2}$$

For $\theta > 0$, we denote by K_θ the θ-envelope of K, i.e.,

$$K_\theta = \bigcup_{x \in K} \bigl(x + \theta\mathbb{U}^n\bigr).$$

We define the following "θ-kernel of K":

$$K_{(-\theta)} = \mathbb{R}^n \setminus \bigl(\mathbb{R}^n \setminus K\bigr)_\theta. \tag{2.3}$$

In the sequel, we assume that in $K_{(-\theta)}$ the number θ is less than the diameter of K. This implies that $K_{(-\theta)}$ is a nonempty set.

Let $f \in B^p_K$. For $\theta > 0$, we say that f is θ-strictly bandlimited to K if $\operatorname{supp} \hat{f} \subset K_{(-\theta)}$. Throughout, the term "$f$ is strictly bandlimited to K" means that f is θ-strictly bandlimited to K for some $\theta > 0$.

We need a few preliminary properties of B^p_K. Let us start with some Nikol'skii's inequalities (see [8, Sect. 3]). Let $f \in B^p_{\mathbb{Q}^n_\sigma}$, where $\mathbb{Q}^n_\sigma = \{x \in \mathbb{R}^n\colon\ |x_k| \leqslant \sigma_k < \infty,\ k = 1, \dots, n\}$. If $1 \leqslant p \leqslant r \leqslant \infty$, then

$$\|f\|_{L^r(\mathbb{R}^n)} \leqslant 2^n \left(\prod_{k=1}^n \sigma_k\right)^{1/p-1/r} \|f\|_{L^p(\mathbb{R}^n)}. \tag{2.4}$$

In the case $1 \leqslant p < \infty$,

$$\lim_{x \to \infty} f(x) = 0. \tag{2.5}$$

Of course, (2.4) and (2.5) still hold for all $f \in B^p_K$ in the case where K is a compact body in $\mathbb{R}^n$. More precisely, given a compact body $K \subset \mathbb{R}^n$ and $1 \leqslant p \leqslant r \leqslant \infty$, there exists a positive constant $A = A(n, p, r, K) < \infty$, depending on n, p, r, and K, such that

$$\|f\|_{L^r(\mathbb{R}^n)} \leqslant A\,\|f\|_{L^p(\mathbb{R}^n)} \tag{2.6}$$

for all $f \in B^p_K$. In other words, the embedding mapping

$$B^p_K \subset B^r_K \tag{2.7}$$

is continuous.

Let p be a 2π-periodic in each variable trigonometric polynomial on $2\pi\mathbb{T}^n$. If $1 \leqslant p \leqslant r \leqslant \infty$, then there exists a constant $c = c(n)$ such that

$$\|p\|_{L^r(2\pi\mathbb{T}^n)} \leqslant c\left(\prod_{k=1}^n m_k\right)^{1/p-1/r} \|p\|_{L^p(2\pi\mathbb{T}^n)}, \tag{2.8}$$

where m_k is the degree of p in the variable x_k (see [8, Sect. 3], [14]).

Recall that the Schwartz class of test functions $S(\mathbb{R}^n)$ consists of the complex-valued infinitely differentiable functions ξ on $\mathbb{R}^n$ satisfying

$$\sup_{x\in\mathbb{R}^n,\,|u|\leqslant k}\left|\left(1+|x|_2\right)^s D^u\xi(x)\right|<\infty \tag{2.9}$$

for all $k, s\in\mathbb{N}$, where u is a nonnegative multiindex, and $|u|=\sum_{j=1}^n u_j$. The dual space $S'(\mathbb{R}^n)$ is the space of tempered distributions. If $\xi\in S(\mathbb{R}^n)$ and $\tau>0$, then the Poisson summation formula gives

$$\sum_{l\in\mathbb{Z}^n}\xi(x+\tau l)=\frac{1}{\tau^n}\sum_{m\in\mathbb{Z}^n}\hat{\xi}\left(\frac{2\pi}{\tau}m\right)\mathrm{e}^{2\pi\mathrm{i}(m,x)/\tau},\quad x\in\mathbb{R}^n \tag{2.10}$$

(see [4, p. 177, (7.2.1)]).

Given a compact set K in $\mathbb{R}^n$, we denote by $S_K^\circ(\mathbb{R}^n)$ the subspace of $S(\mathbb{R}^n)$ consisting of functions strictly bandlimited to K. We will need the following auxiliary statement. For completeness, we also give its proof.

Lemma 1. *Let K be a convex compact body in $\mathbb{R}^n$. If $1\leqslant p<\infty$, then $S_K^\circ(\mathbb{R}^n)$ is an $L^p(\mathbb{R}^n)$-dense subspace of B_K^p.*

Proof. Suppose first that $0\in\operatorname{int}K$. Take any $\zeta\in S(\mathbb{R}^n)$ such that ζ is nonnegative on $\mathbb{R}^n$, $\operatorname{supp}\zeta\subset\mathbb{U}^n$, and

$$\int_{\mathbb{R}^n}\zeta(x)\,\mathrm{d}x=1. \tag{2.11}$$

By the Bochner theorem, the function $\check{\zeta}$ is continuous and positive definite on $\mathbb{R}^n$, and (2.11) implies that

$$\sup_{\mathbb{R}^n}\left|\check{\zeta}(t)\right|\leqslant\check{\zeta}(0)=(2\pi)^{-n}. \tag{2.12}$$

Let $f\in B_K^p$. Given $\delta\in(0,1)$, we set $f_\delta(x)=f(\delta x)$. Let

$$F_\delta(x)=(2\pi)^n f_\delta(x)\check{\zeta}\left(\frac{(1-\delta)r_K}{2}x\right), \tag{2.13}$$

where r_K is defined in (2.1), and $r_K>0$ since $0\in\operatorname{int}K$. We consider $\hat{f}$ as an element of $S'(\mathbb{R}^n)$ with compact support. Then $\hat{f}_\delta*\zeta$ is well defined in $S'(\mathbb{R}^n)$, and

$$\operatorname{supp}\hat{F}_\delta=\operatorname{supp}(\hat{f}_\delta*\zeta)\subset\delta K+\frac{(1-\delta)r_K}{2}\mathbb{U}^n\subset K_{-(\theta)}$$

for all $\theta\in(0,(1-\delta)r_K/2)$. By (2.5), each $f\in B_K^p$, $p\in[1,\infty]$, is a bounded function on $\mathbb{R}^n$. Hence, $F_\delta\in S_K^\circ(\mathbb{R}^n)$.

Let us prove that

$$\lim_{\delta\to1}\|f-F_\delta\|_{L^p(\mathbb{R}^n)}=0. \tag{2.14}$$

To this end, we estimate $\|f-f_\delta\|_{L^p(\mathbb{R}^n)}$ and $\|f_\delta-F_\delta\|_{L^p(\mathbb{R}^n)}$. Set $\varepsilon>0$. Then there exists $a_\varepsilon>0$ such that

$$\|f\|_{L^p(\mathbb{R}^n\setminus a_\varepsilon\mathbb{U}^n)}<\varepsilon. \tag{2.15}$$

Fix any such a_ε. If $1/2 < \delta < 1$, then

$$\|F_\delta\|_{L^p(\mathbb{R}^n\setminus a_\varepsilon\mathbb{U}^n)} \leqslant \|f_\delta\|_{L^p(\mathbb{R}^n\setminus a_\varepsilon\mathbb{U}^n)} < 2^{n/p}\varepsilon \tag{2.16}$$

for all $\delta \in (1/2, 1)$.

Now, we consider the functions f, f_δ, and F_δ on $a_\varepsilon\mathbb{U}^n$. It is known (see, for example, [9]) that if $g \in B_K^p$, $1 \leqslant p \leqslant \infty$, $x, y \in \mathbb{R}^n$, and $\max_{t\in K} |(x-y,t)| \leqslant \pi$, then

$$\big|g(x) - g(y)\big| \leqslant 2\sin\left(\frac{1}{2}\max_{t\in K}\big|(t, x-y)\big|\right)\|g\|_\infty. \tag{2.17}$$

Therefore, if $x \in a_\varepsilon\mathbb{U}^n$ and $1-\delta$ is sufficiently small, then by (2.17), we obtain that

$$\begin{aligned}\|f - f_\delta\|_{L^p(a_\varepsilon\mathbb{U}^n)} &\leqslant 2\big(a_\varepsilon^n m_n\big(\mathbb{U}^n\big)\big)^{1/p}\left|\sin\left(\frac{1-\delta}{2}\max_{t\in K,\, x\in a_\varepsilon\mathbb{U}^n}\big|(x,t)\big|\right)\right| \\ &\leqslant \big(a_\varepsilon^{n+p} m_n\big(\mathbb{U}^n\big)\big)^{1/p}(1-\delta)n^{1/2}R_K < \varepsilon,\end{aligned} \tag{2.18}$$

where R_K is defined in (2.2). In view of (2.12), for $x \in a_\varepsilon\mathbb{U}^n$, we have

$$\begin{aligned}\big|f_\delta(x) - F_\delta(x)\big| &= (2\pi)^n\big|f_\delta(x)\big|\left|\check\zeta(0) - \check\zeta\left(\frac{(1-\delta)r_K}{2}x\right)\right| \leqslant \|f\|_{L^\infty(\mathbb{R}^n)}\int\limits_{\mathbb{R}^n}\big|1 - \mathrm{e}^{\mathrm{i}(1-\delta)r_K(x,t)/2}\big|\zeta(t)\,\mathrm{d}t \\ &\leqslant \max_{t\in\mathbb{U}^n,\, x\in a_\varepsilon\mathbb{U}^n}\big|1 - \mathrm{e}^{\mathrm{i}(1-\delta)r_K(x,t)/2}\big|\|f\|_{L^\infty(\mathbb{R}^n)} \\ &\leqslant \max_{t\in\mathbb{U}^n,\, x\in a_\varepsilon\mathbb{U}^n}\big|(x,t)\big|\frac{(1-\delta)r_K\,\|f\|_{L^\infty(\mathbb{R}^n)}}{\sqrt{2}} \leqslant \left(\frac{n}{2}\right)^{1/2}(1-\delta)r_K a_\varepsilon\|f\|_{L^\infty(\mathbb{R}^n)}.\end{aligned}$$

Hence,

$$\big\|f_\delta(x) - F_\delta(x)\big\|_{L^p(a_\varepsilon\mathbb{U}^n)} \leqslant \big(a_\varepsilon^n m_n\big(\mathbb{U}^n\big)\big)^{1/p}\left(\frac{n}{2}\right)^{1/2}(1-\delta)r_K a_\varepsilon\|f\|_{L^\infty(\mathbb{R}^n)} < \varepsilon \tag{2.19}$$

for sufficiently small $1-\delta$. Since ε is an arbitrary positive number, combining (2.15), (2.16), (2.18), and (2.19), we have (2.14).

In the general case, we take an arbitrary $x_0 \in \operatorname{int} K$. Let $g \in B_K^p$. If

$$f(x) = g(x)\mathrm{e}^{-\mathrm{i}(x,x_0)},$$

then $f \in B^p_{-x_0+K}$, where the body $\operatorname{int}(-x_0 + K)$ contains the origin. For $\delta \in (0,1)$, we set $G_\delta(x) = F_\delta(x)\mathrm{e}^{\mathrm{i}(x,x_0)}$, where F_δ is defined in (2.13). Then (2.14) becomes $\lim_{\delta\to 1}\|G_\delta - g\| = 0$. Finally, each G_δ belongs to $S_K^\circ(\mathbb{R}^n)$, and this completes the proof of the lemma. □

3 PROOFS

For $f \in S(\mathbb{R}^n)$, let

$$E_{f,\tau}(x) := \frac{1}{\tau^n}\sum_{m\in\mathbb{Z}^n\cap\frac{\tau}{2\pi}K}\hat f\left(\frac{2\pi}{\tau}m\right)\mathrm{e}^{2\pi\mathrm{i}(m,x)/\tau}, \quad \tau > 0,\ x \in \mathbb{R}^n. \tag{3.1}$$

We will compare this polynomial with our main trigonometric polynomial (1.4).

Lemma 2. *Let K be a convex compact body in $\mathbb{R}^n$. If $f \in S_K^\circ(\mathbb{R}^n)$ and $p \in [1,\infty)$, then*

$$\lim_{\tau\to\infty} \|f_\tau - E_{f,\tau}\|_{L^p(\tau\mathbb{T}^n)} = 0. \tag{3.2}$$

Proof. First, we claim that there exists $\tau_0 > 0$ such that if $\tau \geqslant \tau_0$, then $E_{f,\tau}$ can be written as

$$E_{f,\tau}(x) = \frac{1}{\tau^n} \sum_{m\in\mathbb{Z}^n\cap n_\tau K} \hat f\Big(\frac{2\pi}{\tau}m\Big)\mathrm{e}^{2\pi\mathrm{i}(m,x)/\tau}, \tag{3.3}$$

where $n_\tau = [\tau/2\pi]_{\mathbb{Z}}$. To that end, it suffices to show that if $\hat f((2\pi/\tau)m_0) \neq 0$ for $m_0 \in \mathbb{Z}^n$, then $m_0 \in n_\tau K$. Since f is θ-strictly bandlimited to K for some $\theta > 0$ and K is convex, we have $(2\pi/\tau)m_0 + \theta\mathbb{U}^n \subset K$. Hence, there exists $\delta_0 > 0$ such that if $\delta \in (0,\delta_0]$, then

$$\frac{2\pi}{\tau}(1+\delta)m_0 \in K. \tag{3.4}$$

Suppose that τ_0 is defined by means of the equality $(1+\delta_0)(1-2\pi/\tau_0) = 1$. Since $1 - 2\pi/\tau < 2\pi n_\tau/\tau \leqslant 1$, we have that for any $\tau \in [\tau_0,\infty)$, there exists $\delta \in (0,\delta_0]$ such that $2\pi n_\tau(1+\delta)/\tau = 1$. Using (3.4), we obtain that $m_0 \in n_\tau K$. This proves (3.3).

Let $\tau \geqslant \tau_0$. Using (3.3), we can estimate $\|f_\tau - E_{f,\tau}\|_{L^p(\tau\mathbb{T}^n)}$ as follows:

$$\begin{aligned}
\|f_\tau - E_{f,\tau}\|_{L^p(\tau\mathbb{T}^n)} &= \frac{1}{\tau^n}\bigg(\int\limits_{\tau\mathbb{T}^n}\bigg|\sum_{m\in\mathbb{Z}^n\cap n_\tau K}\tau^n c_m \mathrm{e}^{2\pi\mathrm{i}(m,x)/\tau} - \sum_{k\in\mathbb{Z}^n\cap\frac{\tau}{2\pi}K}\hat f\Big(\frac{2\pi}{\tau}k\Big)\mathrm{e}^{2\pi\mathrm{i}(k,x)/\tau}\bigg|^p \mathrm{d}x\bigg)^{1/p}\\
&= \frac{1}{\tau^n}\bigg(\int\limits_{\tau\mathbb{T}^n}\bigg|\sum_{m\in\mathbb{Z}^n\cap n_\tau K}\Big(\tau^n c_m - \hat f\Big(\frac{2\pi}{\tau}m\Big)\Big)\mathrm{e}^{2\pi\mathrm{i}(k,x)/\tau}\bigg|^p \mathrm{d}x\bigg)^{1/p}\\
&\leqslant \tau^{n/p-n}\sum_{m\in\mathbb{Z}^n\cap\frac{\tau}{2\pi}K}\Big|\tau^n c_m - \hat f\Big(\frac{2\pi}{\tau}m\Big)\Big|.
\end{aligned} \tag{3.5}$$

Let $s \in \mathbb{N}$. By (2.9) (with $u = 0$), there exists a constant $c(f,s) < \infty$ such that

$$\big|f(x)\big| \leqslant c(f,s)|x|_2^{-s}$$

for $x \in \mathbb{R}^n \setminus \frac{\tau_0}{2}\mathbb{U}^n$. Since $\tau \geqslant \tau_0$, it follows that

$$\begin{aligned}
\Big|\tau^n c_m - \hat f\Big(\frac{2\pi}{\tau}m\Big)\Big| &= \Big|\int\limits_{\mathbb{R}^n\setminus\tau\mathbb{T}^n} f(x)\mathrm{e}^{-2\pi\mathrm{i}(m,x)/\tau}\,\mathrm{d}x\Big| \leqslant \int\limits_{\mathbb{R}^n\setminus\tau\mathbb{T}^n}\big|f(x)\big|\,\mathrm{d}x\\
&\leqslant \int\limits_{\mathbb{R}^n\setminus\frac{\tau}{2}\mathbb{U}^n}\big|f(x)\big|\,\mathrm{d}x \leqslant c(f,s)\int\limits_{\mathbb{R}^n\setminus\frac{\tau}{2}\mathbb{U}^n}|x|_2^{-s}\,\mathrm{d}x.
\end{aligned}$$

If $s > n$, then, via polar coordinates in $\mathbb{R}^n$,

$$\Big|\tau^n c_m - \hat f\Big(\frac{2\pi}{\tau}m\Big)\Big| \leqslant c(f,s)\frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})}\int\limits_{\tau/2}^{\infty}\varrho^{n-s-1}\,\mathrm{d}\varrho = c(f,s)\frac{2\pi^{n/2}}{(s-n)\Gamma(\frac{n}{2})}\tau^{n-s}, \tag{3.6}$$

where Γ is the usual gamma function. Finally, if we choose in (2.9) $u = 0$ and $s > 0$ so that $s > n(1 + 1/p)$, then (3.5) , together with (3.6), proves (3.2) and the lemma. $\square$

Let K be a compact subset of $\mathbb{R}^n$. If $k \in \mathbb{N}$, then

$$D_{k,K}(x) = \sum_{m \in \mathbb{Z}^n \cap kK} \mathrm{e}^{\mathrm{i}(x,m)}$$

is called the Dirichlet kernel of order k related to K. Now, the polynomial (1.4) can be represented as the value of the following integral operator:

$$f_\tau(x) = \frac{1}{\tau^n} \int\limits_{\tau \mathbb{T}^n} f(t) D_{n_\tau,K}\left(\frac{2\pi}{\tau}(x-t)\right) \mathrm{d}t. \tag{3.7}$$

Therefore, setting

$$\Phi_\tau(f) = \varphi_{f,\tau}, \quad \tau \in (0,\infty),$$

we may assume that (1.4) defines on B_K^p the one-parametric family of bounded linear operators $\Phi_\tau : B_K^p \to L^p(\mathbb{R}^n)$. In the next lemma, we give some sufficient conditions under which the family $(\Phi_\tau)_{\tau>0}$ has uniformly bounded norms.

Lemma 3. *Let K be an n-polytope in $\mathbb{R}^n$. If $1 < p < \infty$, then there exists a constant $\varpi(K,p) < \infty$ such that*

$$\big\|\Phi_\tau(f)\big\|_{L^p(\mathbb{R}^n)} \leqslant \varpi(K,p) \|f\|_{L^p(\mathbb{R}^n)} \tag{3.8}$$

for every $f \in B_K^p$ and all $\tau \in (0,\infty)$.

Proof. Given $f \in B_K^p$, we define on $\mathbb{R}^n$ the $2\pi\mathbb{T}^n$-periodic function g_f such that $g_f(v) = f((\tau/2\pi)v)$ for $v \in 2\pi\mathbb{T}^n$. Now, (3.7) implies

$$\begin{aligned} \big\|\Phi_\tau(f)\big\|_{L^p(\mathbb{R}^n)}^p = \big\|\Phi_\tau(f)\big\|_{L^p(\tau\mathbb{T}^n)}^p &= \frac{1}{\tau^{np}} \int\limits_{\tau\mathbb{T}^n} \left| \int\limits_{\tau\mathbb{T}^n} f(t) D_{n_\tau,K}\left(\frac{2\pi}{\tau}(x-t)\right) \mathrm{d}t \right|^p \mathrm{d}x \\ &= \frac{\tau^n}{(2\pi)^{n(p+1)}} \int\limits_{2\pi\mathbb{T}^n} \left| \int\limits_{2\pi\mathbb{T}^n} g_f(v) D_{n_\tau,K}(y-v)\,\mathrm{d}v \right|^p \mathrm{d}y. \end{aligned} \tag{3.9}$$

In [15, Thm. 2.4.5], it is shown that under the conditions of this lemma, there exists a constant $c(K) > 0$, depending only on K, such that

$$\frac{1}{(2\pi)^n} \left(\int\limits_{2\pi\mathbb{T}^n} \left| \int\limits_{2\pi\mathbb{T}^n} h(v) D_{s,K}(y-v)\,\mathrm{d}v \right|^p \mathrm{d}y \right)^{1/p} \leqslant c(K) \left(1 + \frac{4p^2}{p-1}\right)^n \|h\|_{L^p(2\pi\mathbb{T}^n)}$$

for all $s \in \mathbb{N}$ and $h \in L^p(2\pi\mathbb{T}^n)$. Combining this with (3.9), we get

$$\big\|\Phi_\tau(f)\big\|_{L^p(\mathbb{R}^n)} \leqslant c(K) \left(1 + \frac{4p^2}{p-1}\right)^n \|f\|_{L^p(\tau\mathbb{T}^n)}.$$

Thus, we have (3.8) with $\varpi(K,p) = c(K)(1 + 4p^2/(p-1))^n$, and the lemma is proved. $\square$

Proof of Theorem 1. The Banach–Steinhaus theorem, together with Lemmas 1 and 3, implies that it suffices to check (1.7) only for f in $S_K^\circ(\mathbb{R}^n)$. Let $f \in S_K^\circ(\mathbb{R}^n)$. The Poisson summation formula (2.10) gives

$$\sum_{l\in\mathbb{Z}^n} f(x+\tau l) = E_{f,\tau}(x), \quad x \in \mathbb{R}^n, \tag{3.10}$$

where $E_{f,\tau}$ is defined in (3.1). Here the series converges absolutely and uniformly on compact subset of $\mathbb{R}^n$. In particular, the series

$$\sum_{l\in\mathbb{Z}^n\setminus\{0\}} f(x+\tau l)$$

defines on $\tau\mathbb{T}^n$ an $L^p(\tau\mathbb{T}^n)$ function. Since $\lim_{\tau\to\infty}\|f\|_{L^p(\mathbb{R}^n\setminus\tau\mathbb{T}^n)} = 0$, we have from (3.2) and (3.10) that it remains to show that

$$\lim_{\tau\to\infty}\Big\|\sum_{l\in\mathbb{Z}^n\setminus\{0\}} f(x+\tau l)\Big\|_{L^p(\tau\mathbb{T}^n)} = 0. \tag{3.11}$$

Let $s \in \mathbb{N}$. If $x \in \tau\mathbb{T}^n$ and $l \in \mathbb{Z}^n \setminus \{0\}$, then (2.9) implies that there exists $M(s) > 0$ such that

$$\big|f(x+\tau l)\big| \leqslant \frac{M(s)}{(1+|x+\tau l|_2)^s} \leqslant \frac{M(s)}{(1+\frac{\tau}{2}|l|_2)^s} \leqslant \frac{2^s M(s)}{\tau^s |l|_2^s} \leqslant \frac{2^s M(s)}{\tau^s |l|_\infty^s}.$$

We may take $s \in \mathbb{N}$ such that $sp > n$. Then

$$\Big|\sum_{l\in\mathbb{Z}^n\setminus\{0\}} f(x+\tau l)\Big| \leqslant \frac{2^s M(s)}{\tau^s}\sum_{l\in\mathbb{Z}^n\setminus\{0\}}\frac{1}{|l|_\infty^s} \leqslant \frac{2^s M(s)}{\tau^s}\sum_{m=1}^{\infty}\frac{2^n(m-1)^{n-1}}{m^s}.$$

Therefore, exists $M_1(s) > 0$ such that

$$\Big\|\sum_{l\in\mathbb{Z}^n\setminus\{0\}} f(x+\tau l)\Big\|_{L^p(\tau\mathbb{T}^n)} \leqslant M_1(s)\tau^{n/p-s}.$$

This proves (3.11), and the proof of Theorem 1 is complete. □

Proof of Theorem 2. Suppose that $f \in B_K^p$. We claim that

$$M(f) := \sup_{\tau>0}\|f - f_\tau\|_{L^\infty(\mathbb{R}^n)} < \infty. \tag{3.12}$$

In fact, by (2.4), it suffices to show that

$$\sup_{\tau>0}\|f_\tau\|_{L^\infty(\mathbb{R}^n)} < \infty. \tag{3.13}$$

To see this, let $p_\tau(t) := f_\tau(\tau t/2\pi)$, $t \in 2\pi\mathbb{T}^n$. Since $K \subset R_K\mathbb{U}^N \subset R_K\mathbb{Q}^n$, where R_K is defined in (2.2), the polynomial p_τ is 2π-periodic and has degree at most $n_\tau R_K$ in each variable. Now, by (2.8), there exists $c < \infty$ such that

$$\|p_\tau\|_{L^\infty(2\pi\mathbb{T}^n)} \leqslant c(n_\tau R_K)^{n/p}\|p_\tau\|_{L^p(2\pi\mathbb{T}^n)}.$$

Then

$$\|f_\tau\|_{L^\infty(\mathbb{R}^n)} = \|f_\tau\|_{L^\infty(\tau\mathbb{T}^n)} = \|p_\tau\|_{L^\infty(2\pi\mathbb{T}^n)} \leqslant c(n_\tau R_K)^{n/p}\|p_\tau\|_{L^p(2\pi\mathbb{T}^n)}$$
$$= c\left(\frac{2\pi n_\tau}{\tau}R_K\right)^{n/p}\|f_\tau\|_{L^p(\tau\mathbb{T}^n)} \leqslant cR_K^{n/p}\|f_\tau\|_{L^p(\tau\mathbb{T}^n)}.$$

Thus, using Lemma 3, we get

$$\|f_\tau\|_{L^\infty(\mathbb{R}^n)} \leqslant cR_K^{n/p}\big\|\Phi_\tau(f)\big\|_{L^p(\mathbb{R}^n)} \leqslant cR_K^{n/p}\varpi(K,p)\|f\|_{L^p(\mathbb{R}^n)}, \tag{3.14}$$

and this proves (3.13).

In view of (2.5), to prove the theorem, it suffices to show that

$$\lim_{\tau\to\infty}\|f-\varphi_{f,\tau}\|_{L^\infty(\tau\mathbb{T}^n)} = 0.$$

Assume, to the contrary, that there exist $a>0$ and a sequence $\tau_m\to\infty$ such that $\|f-\varphi_{f,\tau_m}\|_{L^\infty(\tau_m\mathbb{T}^n)} \geqslant a$ for all $m\in\mathbb{N}$. Then we may choose a sequence $(x_m)_1^\infty$, where $x_m\in\tau_m\mathbb{T}^n$, so that

$$\big|(f-f_{\tau_m})(x_m)\big| = \big|(f-\varphi_{f,\tau_m})(x_m)\big| \geqslant a. \tag{3.15}$$

We fix $\delta\in(0,1]$ and define

$$\Delta_m = x_m + \frac{\pi\delta}{R_K}\mathbb{U}^n,$$

$m=1,2,\dots$. Suppose that $y\in\Delta_m$. Since $y = x_m + (\pi\delta/R_K)w$, for some $w\in\mathbb{U}^n$, we get

$$\max_{t\in K}\big|(x_m-y,t)\big| = \frac{\pi\delta}{R_K}\max_{t\in K}\big|(w,t)\big| \leqslant \frac{\pi\delta}{R_K}\max_{t\in K}|t|_2 \leqslant \pi\delta \leqslant \pi.$$

Now, using (2.17) and (3.15), we obtain

$$\big|(f-f_{\tau_m})(y)\big| \geqslant \big|(f-f_{\tau_m})(x_m)\big| - 2\|f-f_{\tau_m}\|_{L^\infty(\mathbb{R}^n)}\sin\left(\frac{1}{2}\max_{t\in K}\big|(x_m-y,t)\big|\right)$$
$$\geqslant a - \pi\delta\|f-f_{\tau_m}\|_{L^\infty(\mathbb{R}^n)}. \tag{3.16}$$

Since $a\leqslant M(f)<\infty$, we may take

$$\delta = \frac{a}{2\pi M(f)}.$$

Then (3.16) implies

$$\int\limits_{\tau_m\mathbb{T}^n}\big|(f-f_{\tau_m})(x)\big|^p\,\mathrm{d}x \geqslant \int\limits_{\tau_m\mathbb{T}^n\cap\Delta_m}\big|(f-f_{\tau_m})(y)\big|^p\,\mathrm{d}y \geqslant \left(\frac{a}{2}\right)^p m_n\big(\tau_m\mathbb{T}^n\cap\Delta_m\big). \tag{3.17}$$

We may assume, without loss of generality, that

$$\tau_m \geqslant \frac{2\pi}{\sqrt{n}R_K}$$

for all $m \in \mathbb{N}$. Then

$$m_n\big(\Delta_m \cap \tau_m \mathbb{T}^n\big) \geqslant \frac{\pi\delta\, m_n(\mathbb{U}^n)}{2^n R_K} = \frac{a\, m_n(\mathbb{U}^n)}{2^{n+1} M(f) R_K}. \tag{3.18}$$

Now, combining (3.17) and (3.18), we obtain

$$\int\limits_{\tau_m \mathbb{T}^n} \big|(f - f_{\tau_m})(x)\big|^p \,\mathrm{d}x \geqslant \frac{a^{p+1} m_n(\mathbb{U}^n)}{2^{n+p+1} M(f) R_K}$$

for $m = 1, 2, \ldots$. This contradicts (1.7), and Theorem 2 is proved. □

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